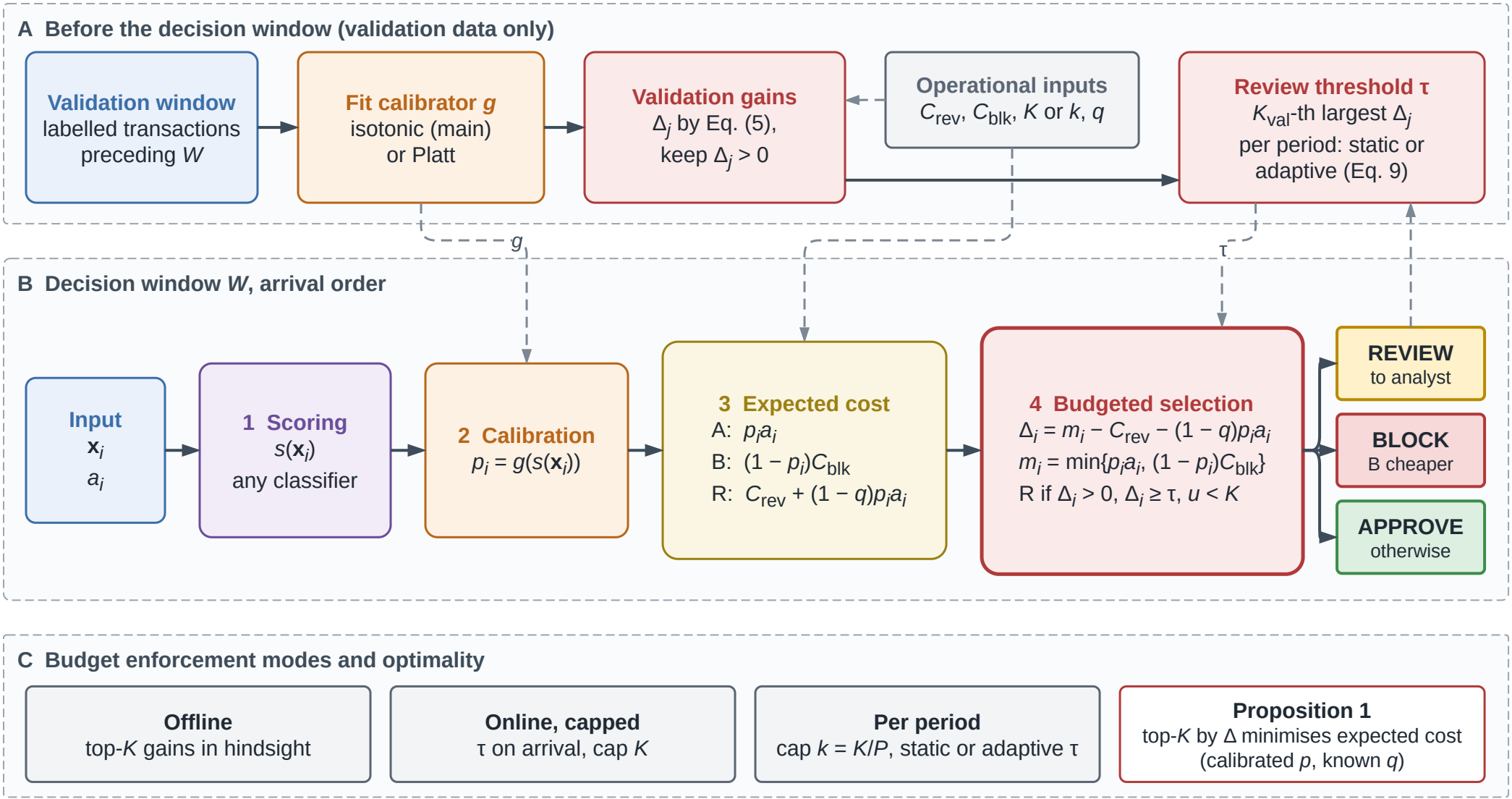

A Before the decision window (validation data only)
Validation window
labelled transactions
preceding W
Fit calibrator g
isotonic (main)
or Platt
Validation gains
Δj by Eq. (5),
keep Δj > 0
Operational inputs
Crev, Cblk, K or k, q
Review threshold τ
Kval-th largest Δj
per period: static or
adaptive (Eq. 9)
g
τ
B Decision window W, arrival order
Input
xi
ai
1 Scoring
s(xi)
any classifier
2 Calibration
pi = g(s(xi))
3 Expected cost
A: piai
B: (1 − pi)Cblk
R: Crev + (1 − q)piai
4 Budgeted selection
Δi = mi − Crev − (1 − q)piai
mi = min{piai, (1 − pi)Cblk}
R if Δi > 0, Δi ≥ τ, u < K
REVIEW
to analyst
BLOCK
B cheaper
APPROVE
otherwise
C Budget enforcement modes and optimality
Offline
top-K gains in hindsight
Online, capped
τ on arrival, cap K
Per period
cap k = K/P, static or adaptive τ
Proposition 1
top-K by Δ minimises expected cost
(calibrated p, known q)